「Galois group of $(x^2-2)(x^2-3)$ over $\mathbb{Q}$.

Since the polynomial $(x^2-2)(x^2-3)$ has four distinct roots $\pm\sqrt{2}, \pm\sqrt{3}$, it is separable.
And, since $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$, $[\mathbb{Q}(\sqrt{2},\sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2},\sqrt{3}), \mathbb{Q}(\sqrt{3})] \cdot [\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] = 4$.
So, $|\mathrm{Aut}(K/\mathbb{Q})| = [K : \mathbb{Q}] = 4$ where $K = \mathbb{Q}(\sqrt{2},\sqrt{3})$, the splitting field over $\mathbb{Q}$.
Since the automorphism leaving $\mathbb{Q}$ fixed must be conjugate the roots,
there are four possible automorphisms;

$$\begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}, \quad \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}, \quad \begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}, \quad \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}.$$

Let $\sigma : K \to K ; a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \mapsto a - b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$
$\tau : K \to K ; a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \mapsto a + b\sqrt{2} - c\sqrt{3} - d\sqrt{6}$.

(That is, $\sigma : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}$ and $\tau : \begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}$, and we can observe that

$$\sigma\tau : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}, \quad \tau\sigma : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}, \text{ so } \sigma^2 = \tau^2 = 1 \text{ and } \sigma\tau = \tau\sigma$$

Now, $\mathrm{Gal}(K/\mathbb{Q}) = \{1, \sigma, \tau, \sigma\tau\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, corresponding σ to $(1,0)$ and τ to $(0,1)$.
Moreover,

$$\begin{tikzcd}
 & \mathbb{Q}(\sqrt{2},\sqrt{3}) = K^{\{id\}} \arrow[dl, no head] \arrow[d, no head] \arrow[dr, no head] & \\
\mathbb{Q}(\sqrt{2}) = K^{\langle\tau\rangle} \arrow[dr, no head] & \mathbb{Q}(\sqrt{6}) = K^{\langle\sigma\tau\rangle} \arrow[d, no head] & \mathbb{Q}(\sqrt{3}) = K^{\langle\sigma\rangle} \arrow[dl, no head] \\
 & \mathbb{Q} = K^{\mathrm{Gal}(K/\mathbb{Q})} &
\end{tikzcd}$$

Note: $\sqrt{2} \in \mathbb{Q}(\sqrt{3})$ because: if $\sqrt{2} = a + b\sqrt{3}$ for some $a, b \in \mathbb{Q}$, then $2 = a^2 + 3b^2 + 2ab\sqrt{3}$.
it is contradiction

「Galois group of x^3-2 over $\mathbb{Q}$.

Denote $\zeta_3 = e^{i\cdot\frac{2\pi}{3}} = \frac{1}{2}(-1+i\cdot\sqrt{3})$, the 3th primitive root of unity.

Since $\zeta_3^3 = 1$, the set $\{\sqrt[3]{2}, \sqrt[3]{2}\,\zeta_3, \sqrt[3]{2}\cdot\zeta_3^2\}$ are all roots of x^3-2 over $\mathbb{Q}$.
So, the splitting field is $K = \mathbb{Q}(\sqrt[3]{2}, \zeta_3) = \mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$.
Observe that: the minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is x^3-2, and
" ζ_3 over $\mathbb{Q}$ is x^2+x+1.
And, since x^2+x+1 is irreducible over $\mathbb{Q}(\sqrt[3]{2})$, (it cannot have root in $\mathbb{Q}(\sqrt[3]{2})$)

$$[\mathbb{Q}(\sqrt[3]{2},\zeta_3):\mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2},\zeta_3):\mathbb{Q}(\sqrt[3]{2})]\cdot[\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}] = 2\cdot 3 = 6.$$

the tower Lemma gave. And, since x^3-2 is separable, $|\mathrm{Aut}(K/\mathbb{Q})| = [K:\mathbb{Q}] = 6$.
Now, Construct explicity the $\mathrm{Aut}(K/\mathbb{Q})$.
Since any automorphism of K leaving $\mathbb{Q}$ fixed must Conjugate the roots,
it must maps $\sqrt[3]{2}$ to $\sqrt[3]{2}$ or $\sqrt[3]{2}\,\zeta_3$ or $\sqrt[3]{2}\cdot\zeta_3^2$
and maps ζ_3 to ζ_3 or ζ_3^2.
So, there are exactly 6 possiblities. Consider two automorphisms

σ satisfying $\sigma(\sqrt[3]{2}) = \sqrt[3]{2}\cdot\zeta_3$ and $\sigma(\zeta_3) = \zeta_3$

and

τ satisfying $\tau(\sqrt[3]{2}) = \sqrt[3]{2}$ and $\tau(\zeta_3) = \zeta_3^2 = -1-\zeta_3$.

Then, $\sigma^3 : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\,\zeta_3 \mapsto \sqrt[3]{2}\cdot\zeta_3^2 \mapsto \sqrt[3]{2}\,\zeta_3^3 = \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3. \end{cases}$, $\tau^2 : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3^2 \mapsto \zeta_3^4 = 1\cdot\zeta_3 = \zeta_3 \end{cases}$

are actually identities. Moreover,

$\sigma^2 : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\cdot\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3 \end{cases}$, $\sigma\tau : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\cdot\zeta_3 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$, $\sigma^2\tau : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\,\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$, $\tau\sigma : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\cdot\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$

So, $\mathrm{Aut}(K/\mathbb{Q}) = \{1, \sigma, \sigma^2, \tau, \sigma\tau, \sigma^2\tau\} \cong S_3$, Corresponding σ to $(1\,2\,3)$, τ to $(1\,2)$
Moreover,

$$\begin{tikzcd}[column sep=small]
 & & \mathbb{Q}(\sqrt[3]{2}, \zeta_3) = K^{\{id\}} \arrow[dll, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\mathbb{Q}(\zeta_3) = K^{\langle\sigma\rangle} \arrow[drr, dash] & & \mathbb{Q}(\sqrt[3]{2}) = K^{\langle\tau\rangle} \arrow[d, dash] & \mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3^2) = K^{\langle\sigma\tau\rangle} \arrow[dl, dash] & \mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3) = K^{\langle\sigma^2\tau\rangle} \arrow[dll, dash] \\
 & & \mathbb{Q} = K^{\langle\sigma,\tau\rangle} & &
\end{tikzcd}$$

How to find a fixed field corresponding to $\langle\sigma\tau\rangle$?
As we know, the $\mathrm{Tr}_{K/K^{\langle\sigma\tau\rangle}}$ is invariant under any automorphism in $\langle\sigma\tau\rangle$.
So, Calculate:

$$\mathrm{Tr}(\sqrt[3]{2}) = id(\sqrt[3]{2}) + (\sigma\circ\tau)(\sqrt[3]{2}) = \sqrt[3]{2} + \sqrt[3]{2}\,\zeta_3 = \sqrt[3]{2}(1+\zeta_3) = \sqrt[3]{2}\cdot(-\zeta_3^2).$$

Therefore, $\sqrt[3]{2}\,\zeta_3^2$ is fixed by id and $\sigma\circ\tau$, so $\mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3^2) \subseteq K^{\langle\sigma\tau\rangle}$.
Moreover, $[\mathbb{Q}(\sqrt[3]{2},\zeta_3) : \mathbb{Q}(\sqrt[3]{2}\cdot\zeta_3^2)] = 2$, so the Galois theorem gives the result.

「Galois group of x^8-2 over $\mathbb{Q}$.

By the eisenstein criterion, $2\in 2\mathbb{Z}$ but $2\notin 2^2\mathbb{Z}$, so x^8-2 is irreducible over $\mathbb{Q}$.
And, the formal derivative $8\cdot x^7$ has only root as zero, but $0^8-2\neq 0$,
So the given polynomial is separable. The polynomial x^8-2 has 8 distinct roots,

$$\sqrt[8]{2}\cdot\zeta_8^k \quad (k=0,1,\cdots,7) \quad \text{where } \zeta_8=\exp\left(2\pi i\cdot\frac{1}{8}\right)=\frac{\sqrt{2}}{2}(1+i)$$

Therefore, the splitting field $K=\mathbb{Q}(\sqrt[8]{2},\cdots,\sqrt[8]{2}\,\zeta_8^7)=\mathbb{Q}(\sqrt[8]{2},i)$, because
the $\subseteq$ inclusion is given by $(\sqrt[8]{2})^4=\sqrt{2}$, so $\zeta_8\in\mathbb{Q}(\sqrt[8]{2},i)$, and
the $\supseteq$ inclusion is given by $\zeta_8^2=\frac{1}{2}(2i)=i\in\mathbb{Q}(\sqrt[8]{2},\cdots,\sqrt[8]{2}\cdot\zeta_8^7)$.
By the irreducibility, the minimal polynomial of $\sqrt[8]{2}$ is x^8-2, and
the minimal polynomial of i is x^2+1 trivially. Since $\mathbb{Q}(\sqrt[8]{2})\subseteq\mathbb{R}$, $i\notin\mathbb{Q}(\sqrt[8]{2})$,
thus the tower lemma gives

$$[K:\mathbb{Q}]=[\mathbb{Q}(\sqrt[8]{2},i):\mathbb{Q}(\sqrt[8]{2})]\cdot[\mathbb{Q}(\sqrt[8]{2}):\mathbb{Q}]=16.$$

Find explicitly $Gal(K/\mathbb{Q})$. Since an automorphism in $Gal(K/\mathbb{Q})$ conjugates the $\sqrt[8]{2}$ and i,
therefore, we can define

$$\sigma:\begin{cases}\sqrt[8]{2}\mapsto\sqrt[8]{2}\cdot\zeta_8\\ i\mapsto i\end{cases} \quad\text{and}\quad \tau:\begin{cases}\sqrt[8]{2}\mapsto\sqrt[8]{2} & \left(\sigma(\zeta_8)=\frac{1+i}{2}\cdot\sigma((\sqrt[8]{2})^4)=\zeta_8^5\right)\\ i\mapsto -i & \left(\tau(\zeta_8)=\tau\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\right)=\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}i=\zeta_8^7\right)\end{cases}$$

Then, just simple calculation gives $id,\sigma,\cdots,\sigma^7,\tau,\tau\sigma,\cdots,\tau\sigma^7$ are mutually distinct. So, $Gal(K/\mathbb{Q})=\{id,\sigma,\cdots,\sigma^7,\tau,\cdots,\tau\sigma^7\}$. Further, observe that

$\sigma^8=\tau^2=1$ and $\sigma\cdot\tau:\begin{cases}\sqrt[8]{2}\mapsto\sqrt[8]{2}\cdot\zeta_8\\ i\mapsto -i\end{cases}$ and

0 1 1+5 1+30

$$\tau\sigma^3:\begin{cases}\sqrt[8]{2}\overset{\sigma}{\mapsto}\sqrt[8]{2}\cdot\zeta_8\overset{\sigma}{\mapsto}\sqrt[8]{2}\cdot\zeta_8^6\overset{\sigma}{\mapsto}\sqrt[8]{2}\cdot\zeta_8^7\overset{\tau}{\mapsto}\sqrt[8]{2}\cdot\zeta_8\\ i\mapsto -i\end{cases}$$

Therefore $Gal(K/\mathbb{Q})=\langle\sigma,\tau\mid\sigma^8=\tau^2=1,\ \sigma\cdot\tau=\tau\sigma^3\rangle$, the quasihedral group.

$$\begin{array}{l} Gal(K/\mathbb{Q})\\ 2\ |\\ \langle\sigma\rangle\\ 2\ |\\ \langle\sigma^2\rangle\\ 2\ |\\ \langle\sigma^4\rangle\\ 2\ |\\ \{id\}\end{array}$$

「Galois group of x^5-3 over $\mathbb{Q}(\zeta)$, where ζ is a primitive fifth root of unity.

Observe that: x^5-3 has five distinct roots, $\sqrt[5]{3}\cdot\zeta^k$ $(k=0,1,\cdots,4)$.
Since the minimal polynomial of ζ over $\mathbb{Q}$ is $x^4+x^3+x^2+x+1$, therefore $[\mathbb{Q}(\zeta):\mathbb{Q}]=4$. And, the minimal polynomial of $\sqrt[5]{3}$ over $\mathbb{Q}$ is x^5-3, therefore $[\mathbb{Q}(\sqrt[5]{3}):\mathbb{Q}]=5$. Therefore, the splitting field of x^5-3 over $\mathbb{Q}$,

$$K=\mathbb{Q}(\sqrt[5]{3},\ \sqrt[5]{3}\zeta^1,\ \sqrt[5]{3}\zeta^2,\ \sqrt[5]{3}\zeta^3,\ \sqrt[5]{3}\zeta^4)=\mathbb{Q}(\sqrt[5]{3},\zeta)=\mathbb{Q}(\sqrt[5]{3})\mathbb{Q}(\zeta)$$

has degree of 20 over $\mathbb{Q}$, that is, $[K:\mathbb{Q}]=20$.
More precisely, by the tower lemma, $[K:\mathbb{Q}]$ is divided by $[\mathbb{Q}(\sqrt[5]{3}):\mathbb{Q}]$ and $[\mathbb{Q}(\zeta):\mathbb{Q}]$.
Since 5 and 4 are relatively prime, $[K:\mathbb{Q}]\geq 20$.
And, $x^5-3\in\mathbb{Q}(\zeta)[x]$ has $\sqrt[5]{3}$ as a root, so

$$[\mathbb{Q}(\sqrt[5]{3},\zeta):\mathbb{Q}(\zeta)]\cdot[\mathbb{Q}(\zeta):\mathbb{Q}]\leq 20.$$

Consequently, $[K:\mathbb{Q}]=20$. Now, by the Fundamental Theorem of Galois Theory, $K/\mathbb{Q}(\zeta)$ is also Galois, with $[K:\mathbb{Q}(\zeta)]=5$.
Therefore, $Gal(K/\mathbb{Q}(\zeta))$ has order of 5, so it must be a cyclic group of order 5. Consequently, the automorphism

$$\sigma:\begin{cases}\sqrt[5]{3}\mapsto\sqrt[5]{3}\cdot\zeta\\ \zeta\mapsto\zeta\end{cases}$$ (It is given by the conjugate isomorphism theorem).

has order of 5, therefore $Gal(K/\mathbb{Q}(\zeta))=\langle\sigma\rangle\cong\mathbb{Z}_5$. ☑

Latticeally,

$$\begin{tikzcd}
& \mathbb{Q}(\sqrt[5]{3},\zeta)=K^{\{id\}} \arrow[dl, dash, "5"'] \arrow[dr, dash, "4"] & \\
\mathbb{Q}(\zeta) = K^{\langle\sigma\rangle} \arrow[dr, dash, "4"'] & & \mathbb{Q}(\sqrt[5]{3}) = K^{\langle\tau\rangle} \arrow[dl, dash, "5"] \\
& \mathbb{Q}=K^{Aut(K/\mathbb{Q})=\langle\sigma,\tau\rangle} &
\end{tikzcd}$$

where $\tau:\begin{cases}\sqrt[5]{3}\mapsto\sqrt[5]{3}\\ \zeta\mapsto\zeta^2.\end{cases}$ Therefore, $Aut(K/\mathbb{Q})\cong\langle(1\ 2\ 3\ 4\ 5),(2\ 3\ 5\ 4)\rangle$.

[Galois group of x^6+3 over $F=\mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Qual, 2020 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $Gal(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.).

1) $F=\mathbb{Q}$.
By the eisenstein creterion, all coefficient of x^6+3 except the leading coefficients are contained in $3\mathbb{Z}$, but the square of constant term $3\notin 3^2\mathbb{Z}$, the given polynomial x^6+3 is irreducible over $\mathbb{Q}$. The x^6+3 has 6 roots,

$$\sqrt[6]{3}\cdot w^{2k+1} \ (k=0,1,\cdots,5) \text{ where } w=\exp\left(\frac{i\pi}{6}\right)=\frac{\sqrt{3}}{2}+i\cdot\frac{1}{2}.$$

Therefore, the splitting field is $K=\mathbb{Q}(\sqrt[6]{3}\cdot w, \sqrt[6]{3}\cdot w^3, \cdots, \sqrt[6]{3}\cdot w^{11})$,
Clearly $\mathbb{Q}(\sqrt[6]{3}\cdot w)\subseteq K$, by the definition of generated field.
Meanwhile, $(\sqrt[6]{3}\cdot w)^3=\sqrt{3}\cdot w^3=\sqrt{3}\cdot\exp(i\pi/2)=\sqrt{3}\cdot i$ and $w^2=\frac{1}{2}+\frac{i\sqrt{3}}{2}$ implies $w^2\in\mathbb{Q}(\sqrt[6]{3}\cdot w)$, so $K=\mathbb{Q}(\sqrt[6]{3}\cdot w)$. Finally, since x^6+3 is irreducible, $[K:\mathbb{Q}]=6$.
Moreover, $Gal(K/\mathbb{Q})\cong S_3$, because: define two automorphisms on K as:
let $\theta=\sqrt[6]{3}\cdot w$ and $\lambda=w^2$. Then, $\theta^3=i\sqrt{3}$ and $\lambda=\frac{1}{2}+\frac{\theta^3}{2}$. Now, let

$$\sigma:\theta\mapsto\theta\cdot\lambda \quad\text{and}\quad \tau:\theta\mapsto\theta\lambda^2.$$

Then, $\sigma(\lambda)=\sigma(\frac{1}{2}+\frac{1}{2}\theta^3)=\frac{1}{2}+\frac{1}{2}\cdot\theta^3\cdot\lambda^3=\frac{1}{2}+\frac{1}{2}\cdot i\sqrt{3}\cdot w^6=\frac{1}{2}(1-i\sqrt{3})=w^{10}=\lambda^{-1}$.
So $\sigma^2(\theta)=\sigma(\theta\lambda)=\theta\lambda\lambda^{-1}=\theta$, it means $\sigma^2=id$.
And $\tau(\lambda)=\frac{1}{2}+\frac{1}{2}\theta^3\cdot\lambda^6=\frac{1}{2}+\frac{1}{2}i\sqrt{3}=\lambda$, so
$\tau^2(\theta)=\tau(\theta\lambda^2)=\theta\lambda^4$, $\tau^3(\theta)=\theta\lambda^6=\theta$, so $\tau^3=id$.
And, $\sigma\tau(\theta)=\sigma(\theta\lambda^2)=\theta\cdot\lambda^{-1}=\theta\cdot\lambda^2$, $(\sigma\tau)^2(\theta)=\sigma\tau(\theta\lambda^2)=\theta\lambda^{-3}=\theta$,
therefore $\sigma\tau\sigma\tau=id$, or $\sigma\tau\sigma=\tau^{-1}$. Now, $Gal(K/F)=\langle\sigma,\tau\rangle\cong D_6\cong S_3$.

Alternatively, if $Gal(K/\mathbb{Q})\cong\mathbb{Z}/6\mathbb{Z}$, the abelian group, then every subgroup is normal.
Therefore every intermidate field E satisfies E/F is Galois. (In other say, normal)
Since $\sqrt[6]{3}\cdot w^{11}\cdot\sqrt[6]{3}\cdot w=\sqrt[3]{3}\in K$, the field $\mathbb{Q}(\sqrt[3]{3})$ is the intermidate field,
but as we know, the minimal polynomial of $\sqrt[3]{3}$ is x^3-3 over $\mathbb{Q}$, but $\mathbb{Q}(\sqrt[3]{3})$ does not have other roots of x^3-3. So, it is contradiction.
By the knowledge in the group theory, there are only two group of order 6, S_3 and $\mathbb{Z}/6\mathbb{Z}$, so $Gal(K/\mathbb{Q})\cong S_3$. ∎

「Galois group of x^6+3 over $F=\mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Qual, 2020 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $\mathrm{Gal}(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.)

2). $F=\mathbb{F}_5$
Suppose that $\alpha\in\overline{\mathbb{F}_5}$ satisfies $\alpha^2=3$. Then, $(\alpha^2)^3+3=30\equiv 0 \bmod 5$,
therefore x^2-3 divides x^6+3, and x^2-3 is irreducible over $\mathbb{F}_5$,
because substituting $x=0,1,\cdots,4$, $x^2-3\neq 0$, by casework. ($\{x^2 \mid x\in\mathbb{F}_5\}=\{0,1,4\}$)
Let $K=\mathbb{F}_5(\alpha)$. Since x^2-3 is the minimal polynomial of α, $[K:\mathbb{F}_5]=2$.
Given $a+b\alpha\in K$ ($a,b\in\mathbb{F}_5$), observe that

$$(a+b\alpha)^6=\underbrace{(a+b\alpha)^5}_{\text{Frobenius automorphism which fixes } \mathbb{F}_5}\cdot(a+b\alpha)=(a+b\alpha^5)\cdot(a+b\alpha)=(a+b\alpha)(a-b\alpha)=a^2\underbrace{+2}_{=-3}b^2=2$$

if and only if:

4 18 22

$$\begin{cases} a=0 \\ b=\pm 1\end{cases} \quad\text{or}\quad \begin{cases} a=\pm 2 \\ b=\pm 2\end{cases} \left(\Leftrightarrow \begin{cases} a=2 \text{ or } 3 \\ b=2 \text{ or } 3\end{cases}\right)$$

So, the six distinct roots are contained in K, so K is splitting field of x^6+3,
and clearly it is separable, therefore $K/\mathbb{F}_5$ is Galois, with the Galois group

$$\mathrm{Gal}(K/\mathbb{F}_5)=\langle x\mapsto x^5\rangle\cong\mathbb{Z}_2.$$

In actually, $K=\mathbb{F}_{5^2}$.

「Galois group of x^6+3 over $F = \mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Qual, 2020 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $Gal(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.).

3). $F = \mathbb{F}_7$

「Galois group of $x^p - x + a$ over $\mathbb{F}_p$ where $a \in \mathbb{F}_p \setminus \{0\}$.

First, we need to show that $x^p - x + a$ is irreducible and separable over $\mathbb{F}_p$.

To show the irreducibility, there are two way;

1). Let $\alpha \in \overline{\mathbb{F}_p}$ be a root of $x^p - x + a$. That is, $\alpha^p - \alpha + a = 0$.

Notice that $(\alpha+1)^p - (\alpha+1) + a = \alpha^p + 1 - (\alpha+1) + a = 0$, because on the field of characteristic p, $x \mapsto x^p$ is Homomorphism. Inductively, $\alpha, \alpha+1, \cdots, \alpha+p-1$ are roots of $x^p - x + a$. Moreover, these are mutually distinct, because $\alpha + i = \alpha + j \iff i = j$ but $\{1, \cdots, p-1\}$ are distinct in the field of characteristic p. Now, over $\overline{\mathbb{F}_p}$,

$$x^p - x + a = (x-\alpha)(x-(\alpha+1))\cdots(x-(\alpha+p-1))$$

If $x^p - x + a$ is reducible over $\mathbb{F}_p$, then $x^p - x + a = f(x)g(x)$ for some $f, g \in \mathbb{F}_p[x]$ and $\deg f, \deg g < p$. By the unique factorization, over $\overline{\mathbb{F}_p}$,

$$f(x) = \prod_{i \in I}(x - (\alpha + i)) \text{ where } I \subsetneq \{0, 1, \cdots, p-1\}.$$

But, then $= x^{|I|} - \left(\sum_{i \in I}(\alpha + i)\right) \cdot x^{|I|-1} + \cdots,$

So $|I| \cdot \alpha + \sum_{i \in I} i \in \mathbb{F}_p$, but then $|I|^{-1} \cdot |I| \cdot \alpha = \alpha \in \mathbb{F}_p$. $(0 < |I| < p)$.

But, $\alpha \in \mathbb{F}_p \iff \alpha$ is a root of $x^p - x \Rightarrow a = 0$, it is contradiction.

Therefore, $x^p - x + a$ is irreducible, and separable because all roots are distinct.

Notice that $\mathbb{F}_p(\alpha)$ is the splitting field of $x^p - x + a$ over $\mathbb{F}$.

Now, $[\mathbb{F}_p(\alpha) : \mathbb{F}_p] = \deg(x^p - x + a) = p$, therefore

$$\mathrm{Gal}(\mathbb{F}_p(\alpha)/\mathbb{F}_p) = \langle x \mapsto x^p \rangle \cong C_p, \text{ the cyclic group of order } p$$

「Galois group of $\mathbb{Q}(\sqrt{2+\sqrt{2}})$ over $\mathbb{Q}$

First, since $x=\sqrt{2+\sqrt{2}} \Rightarrow x^4-4x^2+2=0$, the degree of minimal polynomial of $\sqrt{2+\sqrt{2}}$ divides 4. And, since $1 \notin 2\mathbb{Z}$, $0, 4, 2 \in 2\mathbb{Z}$ but $2 \in 2^2\mathbb{Z}$, so the eisenstein criterion gives that the given polynomial is irreducible, therefore x^4-4x^2+2 is exactly the minimal polynomial of $\sqrt{2+\sqrt{2}}$.

By the fundamental theorem of Algebra, we can factor x^4-4x^2+2 as linear factors, as

$$x^4-4x^2+2=(x^2-(+2+\sqrt{2}))(x^2-(+2-\sqrt{2}))$$
$$=(x-\sqrt{2+\sqrt{2}})(x+\sqrt{2+\sqrt{2}})(x-\sqrt{2-\sqrt{2}})(x+\sqrt{2-\sqrt{2}})$$

To show that $\mathbb{Q}(\sqrt{2+\sqrt{2}})$ is splitting field of x^4-4x^2+2, enough to show that $\sqrt{2-\sqrt{2}} \in \mathbb{Q}(\sqrt{2+\sqrt{2}})$.

observe that: $\sqrt{2}=(\sqrt{2+\sqrt{2}})^2-2$ and

$$(\sqrt{2+\sqrt{2}})\cdot(\sqrt{2-\sqrt{2}})=\sqrt{2}, \text{ so } \sqrt{2-\sqrt{2}}=\frac{\sqrt{2}}{\sqrt{2+\sqrt{2}}}=\sqrt{2+\sqrt{2}}-\frac{2}{\sqrt{2+\sqrt{2}}}$$

therefore $\sqrt{2-\sqrt{2}} \in \mathbb{Q}(\sqrt{2+\sqrt{2}})$. And, the formal derivative $4x^3-8x=4x(x^2-2)$ has roots as $0, \pm\sqrt{2}$, so that x^4-4x^2+2 is separable.

This proves $\mathbb{Q}(\sqrt{2+\sqrt{2}})/\mathbb{Q}$ is Galois, and has degree of 4.

Now, consider an automorphism on $\mathbb{Q}(\sqrt{2+\sqrt{2}})$ as

$$\sigma: \underbrace{\sqrt{2+\sqrt{2}}}_{\alpha} \mapsto \underbrace{\sqrt{2-\sqrt{2}}}_{\beta}.$$

$\alpha^4-4\alpha^2+2=0$ / $\beta=$

$\alpha^3-4\alpha=-\frac{2}{\alpha}$

$\beta^4-4\beta^2=-2$

$-\frac{2}{\beta}+\beta$

Then, $\beta=\alpha^3-3\alpha$, so

$$\sigma^2(\alpha)=\sigma(\beta)=\sigma(\alpha)^3-3\sigma(\alpha)=\beta^3-3\beta=\beta-\frac{2}{\beta}=-\alpha$$

So σ has order 4, therefore $\mathrm{Gal}((\mathbb{Q}(\sqrt{2+\sqrt{2}})/\mathbb{Q})=\langle\sigma\rangle \cong \mathbb{Z}/4\mathbb{Z}$.

$\mathbb{Q}(\sqrt{2+\sqrt{2}})$
|
$\mathbb{Q}(\sqrt{2})$ fixed by $\langle\sigma^2\rangle$
|
$\mathbb{Q}$

Note: $\cos 2\alpha=\cos^2\alpha-\sin^2\alpha=\cos^2\alpha+\cos^2\alpha-1 \Rightarrow \cos^2\alpha=\frac{1+\cos 2\alpha}{2}=1-\sin^2\alpha$.

So, $\cos\frac{\pi}{8}=\sqrt{\frac{1}{2}+\frac{1}{2}\cdot\frac{\sqrt{2}}{2}}=\frac{1}{2}\sqrt{2+\sqrt{2}}$ and $\sin\frac{\pi}{8}=\frac{1}{2}\sqrt{2-\sqrt{2}}$

Biquadratic criterion: $2^2-2=2$ is not square.

Galois group of $\mathbb{Q}(\sqrt{3+\sqrt{3}})$ over $\mathbb{Q}$

Galois group of $\mathbb{Q}(\sqrt{p}+\sqrt{q})$ over $\mathbb{Q}$, where p and q are distinct prime (UCLA, Spring 2023)

First, $[\mathbb{Q}(\sqrt{p}):\mathbb{Q}]=2$. because x^2-p has $\sqrt{p}$ as root and it is irreducible, since x^2-p has no root in $\mathbb{Q}$.

And, $[\mathbb{Q}(\sqrt{p},\sqrt{q}):\mathbb{Q}(\sqrt{p})]=2$, because x^2-q has $\sqrt{q}$ as root and $\sqrt{q}\notin\mathbb{Q}(\sqrt{p})$, since: if $\sqrt{q}\in\mathbb{Q}(\sqrt{p})$, then $\sqrt{q}=a+b\sqrt{p}$, it implies

$$q=a^2+b^2p+2ab\sqrt{p} \Rightarrow \sqrt{p}=\frac{a^2+b^2p}{2ab} \quad (a,b\neq 0)$$

Since $\sqrt{p}\notin\mathbb{Q}$, it is contradiction.

Now, to show $\mathbb{Q}(\sqrt{p}+\sqrt{q})=\mathbb{Q}(\sqrt{p},\sqrt{q})$, observe that

$$\sqrt{p}+\sqrt{q}\in\mathbb{Q}(\sqrt{p},\sqrt{q}) \Rightarrow \mathbb{Q}(\sqrt{p}+\sqrt{q})\subseteq\mathbb{Q}(\sqrt{p},\sqrt{q}).$$

By the tower lemma, $[\mathbb{Q}(\sqrt{p},\sqrt{q}):\mathbb{Q}]=[\mathbb{Q}(\sqrt{p},\sqrt{q}):\mathbb{Q}(\sqrt{p})][\mathbb{Q}(\sqrt{p}):\mathbb{Q}]=4$, enough to show that $[\mathbb{Q}(\sqrt{p}+\sqrt{q}):\mathbb{Q}]=4$.

If $[\mathbb{Q}(\sqrt{p}+\sqrt{q}):\mathbb{Q}]=1$, then $\sqrt{p}+\sqrt{q}\in\mathbb{Q}(\sqrt{p}+\sqrt{q})=\mathbb{Q}$, it is contradiction.

If $[\mathbb{Q}(\sqrt{p}+\sqrt{q}):\mathbb{Q}]=3$, then

$$4=[\mathbb{Q}(\sqrt{p},\sqrt{q}):\mathbb{Q}(\sqrt{p}+\sqrt{q})][\mathbb{Q}(\sqrt{p}+\sqrt{q}):\mathbb{Q}]=n\cdot 3 \text{ for some } n\in\mathbb{N},$$

but trivially $3\nmid 4$, it contradicts

If $[\mathbb{Q}(\sqrt{p}+\sqrt{q}):\mathbb{Q}]=2$, then $\sqrt{p}+\sqrt{q}$ is a root of degree 2 polynomial over $\mathbb{Q}$, that is, $(\sqrt{p}+\sqrt{q})^2+a(\sqrt{p}+\sqrt{q})+b=0$ for some $a,b\in\mathbb{Q}$,

$$=\sqrt{pq}+\sqrt{p}+\sqrt{q}+p+q+b=0$$ a는 어디에 팔아먹었냐?

$$\Rightarrow pq=(\sqrt{p}+\sqrt{q})^2+(p+q+b)^2=(p+q+b)^2+p+q+2\sqrt{pq}$$

$$\Rightarrow \sqrt{pq}\in\mathbb{Q}$$

But it is contradiction, so, $\mathbb{Q}(\sqrt{p}+\sqrt{q})=\mathbb{Q}(\sqrt{p},\sqrt{q})$.

Clearly, $\mathbb{Q}(\sqrt{p},\sqrt{q})$ is splitting field of $(x^2-p)(x^2-q)$, $\mathbb{Q}(\sqrt{p}+\sqrt{q})/\mathbb{Q}$ is Galois, and define

$$\sigma:\begin{cases}\sqrt{p}\mapsto-\sqrt{p}\\ \sqrt{q}\mapsto\sqrt{q}\end{cases} \quad\text{and}\quad \tau:\begin{cases}\sqrt{p}\mapsto\sqrt{p}\\ \sqrt{q}\mapsto-\sqrt{q}.\end{cases}$$

These are well-defined automorphisms of $\mathbb{Q}(\sqrt{p}+\sqrt{q})$ by the Galois theorem.

Moreover, $\mathrm{Gal}(\mathbb{Q}(\sqrt{p}+\sqrt{q})/\mathbb{Q})=\langle\sigma,\tau\rangle\cong\mathbb{Z}_2\times\mathbb{Z}_2$. ▨

[Galois group of $\mathbb{Q}(\sqrt{p}+\sqrt{q})$ over $\mathbb{Q}$, where p and q are distinct prime] Revised.

First, $[\mathbb{Q}(\sqrt{p}):\mathbb{Q}]=2$, because x^2-p is irreducible over $\mathbb{Q}$, since x^2-p has no root in $\mathbb{Q}$. And, $[\mathbb{Q}(\sqrt{p},\sqrt{q}):\mathbb{Q}(\sqrt{p})]=2$, because x^2-q has no root in $\mathbb{Q}(\sqrt{p})$. Specifically, if x^2-q has a root in $\mathbb{Q}(\sqrt{p})$, that is, $\sqrt{q}\in\mathbb{Q}(\sqrt{p})$, then $\sqrt{q}=a+b\sqrt{p}$ for some $a,b\in\mathbb{Q}$.

It implies
$$q=a^2+b^2p+2ab\sqrt{p},$$

but since $\sqrt{p}\in\mathbb{Q}$, either a or b must be zero.

But, $a=0$ means $\sqrt{q}=b\sqrt{p} \Rightarrow q=b^2p$ is not prime when $b\neq 1$, and $q=p$ is contradiction to q and p are distinct, so $b=1$ false.

And, $b=0$ means $\sqrt{q}=a\in\mathbb{Q}$, contradiction. Therefore, by the tower lemma,
$$[\mathbb{Q}(\sqrt{p},\sqrt{q}):\mathbb{Q}]=[\mathbb{Q}(\sqrt{p},\sqrt{q}):\mathbb{Q}(\sqrt{p})][\mathbb{Q}(\sqrt{p}):\mathbb{Q}]=4.$$

To show $\mathbb{Q}(\sqrt{p}+\sqrt{q})=\mathbb{Q}(\sqrt{p},\sqrt{q})$, observe that
$$\sqrt{p}+\sqrt{q}\in\mathbb{Q}(\sqrt{p},\sqrt{q}) \Rightarrow \mathbb{Q}(\sqrt{p}+\sqrt{q})\subseteq\mathbb{Q}(\sqrt{p},\sqrt{q})$$

And, observe that
$$p-q=\sqrt{p}^2-\sqrt{q}^2=(\sqrt{p}+\sqrt{q})(\sqrt{p}-\sqrt{q}) \Rightarrow \sqrt{p}-\sqrt{q}=\frac{p-q}{\sqrt{p}+\sqrt{q}}\in\mathbb{Q}(\sqrt{p}+\sqrt{q})$$

so, $\sqrt{p}=\frac{1}{2}(\sqrt{p}+\sqrt{q}+\sqrt{p}-\sqrt{q})$ and $\sqrt{q}=\frac{1}{2}(\sqrt{p}+\sqrt{q}-(\sqrt{p}-\sqrt{q}))$

are contained in $\mathbb{Q}(\sqrt{p}+\sqrt{q})$. Thus $\mathbb{Q}(\sqrt{p}+\sqrt{q})=\mathbb{Q}(\sqrt{p},\sqrt{q})$.

And, $\mathbb{Q}(\sqrt{p},\sqrt{q})$ is the splitting field of $(x^2-p)(x^2-q)$ over $\mathbb{Q}$,

Therefore, $\mathbb{Q}(\sqrt{p},\sqrt{q})/\mathbb{Q}$ is Galois, so it has four automorphism that fixes $\mathbb{Q}$.

Now, we can explicit two automorphisms:
$$\sigma:\begin{cases}\sqrt{p}\mapsto\sqrt{p}\\ \sqrt{q}\mapsto-\sqrt{q}\end{cases} \quad\text{and}\quad \tau:\begin{cases}\sqrt{p}\mapsto-\sqrt{p}\\ \sqrt{q}\mapsto\sqrt{q}\end{cases}$$

Then, $\sigma^2=\tau^2=(\sigma\tau)^2=id$ by just simple calculation.

Consequently, $Gal(\mathbb{Q}(\sqrt{p},\sqrt{q})/\mathbb{Q})=\langle\sigma,\tau\rangle\cong\mathbb{Z}_2\times\mathbb{Z}_2$,

$$\begin{tikzcd}
 & \mathbb{Q}(\sqrt{p}+\sqrt{q})=K=\mathbb{Q}(\sqrt{p},\sqrt{q}) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt{p}) = K^{\langle\sigma\rangle} \arrow[dr, dash] & \mathbb{Q}(\sqrt{pq}) = K^{\langle\sigma\tau\rangle} \arrow[d, dash] & \mathbb{Q}(\sqrt{q}) = K^{\langle\tau\rangle} \arrow[dl, dash] \\
 & \mathbb{Q}. &
\end{tikzcd}$$

「Galois group of $\mathbb{Q}(x)/\mathbb{Q}(x^2)$

「Galois group of $\mathbb{Q}(x,y)/\mathbb{Q}(xy)$